Karanveer Singh

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Available for co-op starting Jan. 2027

Education

Simon Fraser University Burnaby, BC, Canada
Bachelor of Science in Computing Science

Experience

Founder & Software Developer May 2026 – Present
CallCatch (Self-Founded SaaS) Surrey, BC

- Built and launched a missed-call recovery platform for trades and dental offices that texts every missed caller back in under 5 seconds, capturing leads that would otherwise hit voicemail and call a competitor (Next.js, Supabase, Twilio, Groq LLM)
- Engineered an LLM triage pipeline that collects caller name, address, and issue over SMS and auto-escalates emergencies (flooding, burst pipes) ahead of routine requests so urgent jobs reach the owner first
- Running in production on live calls for a local trades business and backed by 40 automated tests; own architecture, testing, pricing, and outreach end to end

Operations & Technical Lead Aug. 2026
The J Healthcare Initiative (Riipen Labs) Remote

- Led operations and technical scope on a 5-person team in a paid client engagement, auditing a harm-reduction nonprofit's branding and messaging across its website, Instagram, and LinkedIn
- Prototyped a redesigned homepage in HTML/CSS with a unified color system, clearer navigation, and a "find your path" section routing new volunteers, interns, and partners to the right next step
- Wrote the implementation plan and impact analysis for the final client deck, merging two competing recommendations into one sequenced rollout a team with no marketing staff could maintain

Projects

AutoCurve Assistant | *Python, FastAPI, JavaScript, scikit-learn, OpenRouter* Jan. 2026

- Predicted used-vehicle prices from 52K+ listings, with the model explaining 84% of price variance (R^2=0.84) on held-out data
- Built a FastAPI REST backend and JavaScript frontend for real-time vehicle valuations, integrating structured vehicle data with model predictions
- Integrated LLM-based image condition scoring via OpenRouter to quantify vehicle condition and adjust price estimates

MarketPulse | *Next.js, TypeScript, React, MongoDB, Finnhub, Inngest, Gemini* May 2026

- Built a full-stack stock dashboard with authenticated accounts, watchlists, an investment tracker with live profit/loss calculation, and price alerts, backed by MongoDB and scoped per user
- Engineered 3 Inngest background workflows, including a price-alert job polling Finnhub every 15 minutes to trigger above/below target alerts and a daily market-summary email pipeline
- Integrated Gemini for AI-generated market summaries and welcome emails with safe fallbacks, and hardened the app with server-side validation, security headers, and Upstash Redis rate limiting

Myopia Progression Predictor | *Python, scikit-learn, pandas, Streamlit* Dec. 2025 – Mar. 2026

- Built an end-to-end ML pipeline predicting long-term myopia progression from lifestyle and patient data, reaching 0.40 diopter MAE on held-out data
- Engineered domain-specific features to capture non-linear effects of risk factors on progression
- Deployed the model as an interactive Streamlit app that returns real-time progression estimates from user inputs

Technical Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, FastAPI, Node.js, Streamlit, scikit-learn, pandas, NumPy
AI/ML: LLM integration, prompt engineering, regression modeling, feature engineering, model evaluation
APIs & Integrations: REST APIs, Twilio, Groq, OpenRouter, Gemini, Finnhub, Inngest
Databases: PostgreSQL, Supabase, MongoDB, Redis
Developer Tools: Git, GitHub, Vercel, JUnit, JaCoCo